II. End User Access Mechanism

The end users of the system (Security Analysts and System Administrators) can access application services through multiple secure mechanisms. The system supports web-based access, REST API interaction, and role-based access control to ensure efficient and controlled operations.

1. Web-Based Interface (Frontend – Planned Enhancement)

- A centralized web dashboard will be provided for monitoring and managing security incidents.
- Users can view detected incidents in real time.
- Users can check incident status (Open, Escalated, Resolved).
- Users can approve or reject automated response actions.
- The frontend communicates with backend services using REST APIs.

2. REST API Access

- All system components expose REST API endpoints.
- Users can interact using Swagger UI through a web browser.
- API testing tools such as Postman or Thunder Client can be used.
- Command-line tools such as curl are supported.

Example Endpoints:

- /ingest – Submit raw logs
- /incident – View or manage incidents
- /respond – Trigger or manage response decisions

3. Role-Based Access Control

Role	Permissions
Security Analyst	View incidents, monitor status, approve/reject automated actions
System Administrator	Configure playbooks, define policies, manage system settings

4. Interaction Flow

```
End User (Security Analyst)
                ↓
Web Dashboard / API Client
                ↓
   Log Ingestion Service
                ↓
  Log Processing Service
                ↓
     Detection Service
                ↓
Incident Management Service
                ↓
Response Automation Engine
```